Project Charter: Data-Driven Media Optimization for E3 Media Crafters Pvt Ltd

1. Project Title

Data-Driven Media Optimization for E3 Media Crafters Pvt Ltd

2. Project Sponsor

E3 Media Crafters Pvt Ltd
A cutting-edge digital media and creative agency focused on innovative marketing and branding solutions.

3. Project Manager

Mr. Ajit Tripathi

4. Project Overview

E3 Media Crafters Pvt Ltd seeks to enhance its marketing and branding strategies by leveraging data science to optimize media campaigns. The project aims to develop a system that analyzes customer feedback, predicts campaign performance trends, and classifies visual content to improve audience targeting and engagement. Using the Yelp Open Dataset, the system will integrate Python, Deep Learning for Natural Language Processing (NLP), Time Series Analysis (TSA), and Computer Vision (CV), with visualizations in Power BI and an interactive interface via Streamlit. The project will follow the CRISP-DM framework and be executed within an 8-week internship program.

Business Problem

E3 Media Crafters needs to:

- Understand customer sentiment from feedback (e.g., reviews, social media).
- Forecast campaign performance to allocate resources effectively.
- Analyze visual content to identify attributes that drive engagement.

Project Objectives

- Develop an NLP model to classify sentiment from customer feedback.
- Use TSA to predict future campaign performance trends.
- Apply CV to classify images associated with campaigns.
- Create Power BI dashboards for stakeholder insights.
- Build a Streamlit app for interactive analysis.

Expected Outcomes

- Actionable insights to optimize media campaigns.
- Enhanced customer engagement through targeted strategies.
- A prototype system scalable for future use.

5. Project Scope

In Scope

- Data preprocessing of customer feedback (text), campaign performance data (time-stamped), and images.
- Development of deep learning models for:
    - Sentiment analysis (NLP).
    - Campaign performance forecasting (TSA).
    - Image classification (CV).
- Use of Python, Power BI, and Streamlit.
- CRISP-DM framework execution.

Out of Scope

- Real-time data integration.
- Audio or advanced video analysis.
- Production-level deployment.

6. Deliverables

1. **Project Charter**: Outlining objectives, scope, and timeline.
2. **Data Dictionary**: Key dataset fields (e.g., review_id, text, date, photo_id).
3. **EDA Report**: Visualizations of sentiment, trends, and images.
4. **Cleaned Datasets**:
    - NLP: Preprocessed text with sentiment labels.
    - TSA: Time-stamped performance data.
    - CV: Labeled images.
5. **Deep Learning Models**:
    - NLP: Sentiment classifier (e.g., BERT).
    - TSA: Forecasting model (e.g., LSTM).
    - CV: Image classifier (e.g., CNN).
6. **Evaluation Reports**:
    - NLP: Accuracy, precision, recall.
    - TSA: MAE, RMSE.

    - CV: Confusion matrix, accuracy.
7. **Power BI Dashboard**: Visualizing sentiment, forecasts, and image insights.
8. **Streamlit App**: For text input, forecast viewing, and image analysis.
9. **Final Presentation**: Summarizing findings (15-20 minutes).
10. **Final Report**: Detailed project documentation (10-15 pages).

7. Timeline and Milestones

Week	CRISP-DM Phase	Tasks	Deliver
1	Business Understanding	Define problem, research needs	Project charter
2	Data Understanding	Explore Yelp dataset, perform EDA	Data dictionary, EDA report
3	Data Preparation	Preprocess text and time series data	Cleaned NLP and TSA datas
4	Data Preparation	Process image data	Cleaned CV dataset
5	Modeling	Train NLP model	NLP model, evaluation repor
6	Modeling	Train TSA model	TSA model, evaluation repor
7	Modeling	Train CV model	CV model, evaluation report
8	Evaluation & Deployment	Create Power BI dashboard, Streamlit app, finalize deliverables	Power BI dashboard, Stream

Key Milestones

- **Week 2**: Business and data understanding completed.
- **Week 4**: Datasets prepared.
- **Week 6**: Models trained.
- **Week 8**: Solution deployed, deliverables submitted.

8. Roles and Responsibilities

- **Project Sponsor (E3 Media Crafters Pvt Ltd)**:
    - Defines business requirements.
    - Approves charter and deliverables.
- **Project Manager**:
    - Oversees execution.
    - Conducts weekly check-ins.
- **Interns**:
    - Work independently.
    - Deliver tasks and collaborate.

9. Datasets

Primary dataset: **Yelp Open Dataset**

[https://business.yelp.com/data/resources/open-dataset/\]

- **Reviews**: For NLP (sentiment analysis).
- **Timestamps**: For TSA (forecasting).
- **Photos**: For CV (attribute classification).

10. Technologies

- **Python**: Data processing, modeling.
- **Deep Learning**:
  - NLP: BERT/LSTM.
  - TSA: LSTM/ARIMA.
  - CV: CNNs.
- **Power BI**: Dashboards.
- **Streamlit**: Interactive app.
- **Libraries**: Pandas, NumPy, TensorFlow/Keras, NLTK, OpenCV.

11. Risks and Mitigations

Risk	Mitigation
Limited computational resources	Use Google Colab or AWS.
Data quality issues	Apply data augmentation, SMOTE.
Integration challenges	Use CSV files, clear APIs.

12. Communication Plan

- **Weekly Check-Ins**: Fridays, 30 minutes.
- **Milestone Reviews**: Deliverables due Sunday, feedback by Tuesday.
- **Final Presentation**: Week 8, to stakeholders.

13. Approval

- **Project Sponsor**: [Awaiting Signature]
- **Project Manager**: [Awaiting Signature]
- **Date**: [To be confirmed]

Detailed Analysis: Data-Driven Media Optimization Project

Background

E3 Media Crafters Pvt Ltd, a digital media agency, aims to leverage data science to enhance its marketing strategies. The project uses the Yelp Open Dataset to simulate real-world media optimization, analyzing customer feedback, predicting campaign trends, and classifying visual content. The CRISP-DM framework ensures a structured approach, while the 8-week timeline aligns with an internship program.

Project Rationale

The evidence suggests that data-driven insights can significantly improve campaign performance. Sentiment analysis helps understand audience perceptions, time series forecasting optimizes resource allocation, and image classification identifies engaging visual elements. Integrating Python, Deep Learning, Power BI, and Streamlit creates a comprehensive solution tailored to E3's needs.

Dataset Utilization

The Yelp Open Dataset provides:

- **Reviews**: Over 6.9 million entries for NLP tasks.
- **Timestamps**: For TSA, enabling trend analysis.
- **Photos**: Over 200,000 images with attributes like "ambience_is_classy" for CV.

Dataset Suitability

Component	Dataset Feature	Task	Suitability
NLP	Review text, star ratings	Sentiment analysis	High, supports labeling and modeling.
TSA	Review dates	Performance forecasting	High, enables time series creation.
CV	Photos, attributes	Image classification	High, provides labeled images.

Task Breakdown

Week 1: Business Understanding

- Research E3's marketing challenges.
- Deliverable: Project charter.

Week 2: Data Understanding

- Explore Yelp dataset.
- Deliverables: Data dictionary, EDA report with visualizations (e.g., sentiment distribution).

Week 3-4: Data Preparation

- Clean text (tokenization, stop words).
- Aggregate time series data.
- Process images (resize, normalize).
- Deliverables: Cleaned datasets.

Week 5: NLP Modeling

- Train BERT or LSTM for sentiment.
- Deliverables: Model, evaluation report (accuracy ~85% expected).

Week 6: TSA Modeling

- Train LSTM for forecasting.
- Deliverables: Model, forecast plots, metrics (MAE < 0.1 targeted).

Week 7: CV Modeling

- Train CNN for image attributes.
- Deliverables: Model, confusion matrix (accuracy ~80% expected).

Week 8: Deployment

- Create Power BI dashboards (sentiment trends, forecasts).
- Build Streamlit app for user interaction.
- Deliverables: Dashboard, app, presentation, report.

Technology Integration

- **Python**: Core for data processing (Pandas) and modeling (TensorFlow).
- **Deep Learning**:
    - NLP: BERT for high accuracy.
    - TSA: LSTM for sequential data.
    - CV: CNNs for image tasks.
- **Power BI**: Visualizes insights for stakeholders.
- **Streamlit**: Enables real-time interaction.

Risk Management

- **Resource Constraints**: Cloud platforms mitigate training demands.
- **Data Issues**: Preprocessing ensures quality.
- **Timeline Risks**: Weekly check-ins keep progress on track.

Timeline Feasibility

The 8-week schedule is tight but feasible:

- Weeks 1-2 establish foundations.
- Weeks 3-4 focus on data readiness.
- Weeks 5-7 allow iterative modeling.
- Week 8 ensures polished deliverables.

Deliverable Details

- **EDA Report**: Includes histograms, time series plots.
- **Models**: Saved as .pkl files for reuse.

- **Power BI Dashboard**: Interactive, with filters for business categories.
- **Streamlit App**: User-friendly, with input fields and visual outputs.

Conclusion

This project equips E3 Media Crafters with a robust system to optimize campaigns, leveraging cutting-edge data science techniques. The Yelp dataset's versatility ensures all objectives are met, while the CRISP-DM framework guarantees a professional outcome.

Key Citations

- Yelp Open Dataset for Research and Analysis
- CRISP-DM Framework for Data Mining Projects